8. LINUX FILE SYSTEM ANALYSIS

AIM

To analyze the Linux file system and understand how files, directories, and permissions are structured and used for system management and security.

PROCEDURE

1. Study the structure of the Linux file system.
2. Understand important directories such as /etc, /var, /home, and /root.
3. Analyze file permissions, ownership, and access controls.
4. Use Linux commands to navigate and inspect files and directories.
5. Identify suspicious files and potential security risks in the system.

QUESTIONS

TASK 1 - Investigation Setup

After updating the `PATH` and `LD_LIBRARY_PATH` environment variables, run the command `check-env`. What is the flag that is returned in the output?

THM{5514ec4f1ce82f63867806d3cd95dbd8}

✓ Correct Answer

TASK 2 - Files, Permissions, and Timestamps

To practice your skills with the `find` command, locate all the files that the user **bob** created in the past 1 minute. Once found, review its contents. What is the flag you receive?

THM{0b1313afd2136ca0faafb2daa2b430f3}

✓ Correct Answer

Extract the metadata from the `reverse.elf` file. What is the file's MIME type?

application/octet-stream

✓ Correct Answer

Run the `stat` command against the `/etc/hosts` file on the compromised web server. What is the full **Modify Timestamp (mtime)** value?

2020-10-26 21:10:44.000000000 +0000

✓ Correct Answer

TASK 3 - Users and Groups

Investigate the user accounts on the system. What is the name of the backdoor account that the attacker created?

b4ckd00r3d — ✓ Correct Answer

What is the name of the group with the group ID of **46**?

plugdev — ✓ Correct Answer

View the `/etc/sudoers` file on the compromised system. What is the full path of the binary that Jane can run as sudo?

/usr/bin/pstree — ✓ Correct Answer

TASK 4 - User Directories and Files

View Jane's `.bash_history` file. What flag do you see in the output?

THM{f38279ab9c6af1215815e5f7bbad891b} — ✓ Correct Answer

What is the hidden flag in Bob's home directory?

THM{6ed90e00e4fb7945bead8cd59e9fcd7f} — ✓ Correct Answer

Run the `stat` command on Jane's `authorized_keys` file. What is the full timestamp of the most recent modification?

2024-02-13 00:34:16.005897449 +0000 — ✓ Correct Answer

TASK 5 - Binaries and Executables

Run the `debsums` utility on the compromised host to check only configuration files. Which file came back as altered?

/etc/sudoers — ✓ Correct Answer

What is the `md5sum` of the binary that the attacker created to escalate privileges to root?

7063c3930affe123baecd3b340f1ad2c — ✓ Correct Answer

TASK 6 - Rootkits

Run *chkrootkit* on the affected system. What is the full path of the `.sh` file that was detected?

/var/tmp/findme.sh — ✓ Correct Answer

Run *rkhunter* on the affected system. What is the result of the `(UID 0) accounts` check?

Warning — ✓ Correct Answer

RESULT

Successfully analyzed the Linux file system and understood how file structure, permissions, and commands are used to manage and secure a Linux system.